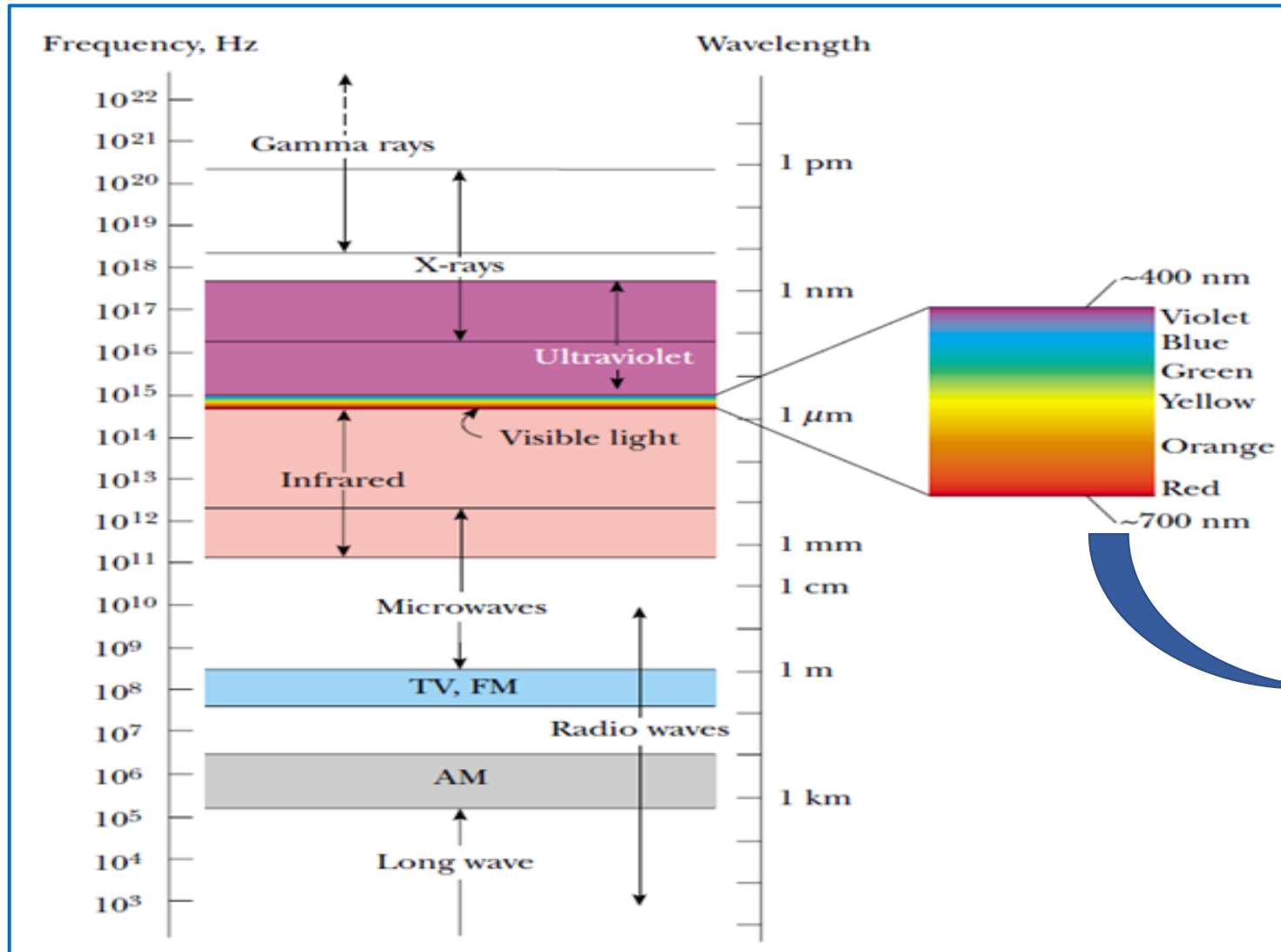


Electromagnetic spectrum



The expanded visible spectrum

Note the overlap between adjacent wave types.

Optical Properties of Materials

It describes a material behavior when light is incident on its surface, or how a material interacts with the incident electromagnetic radiation.

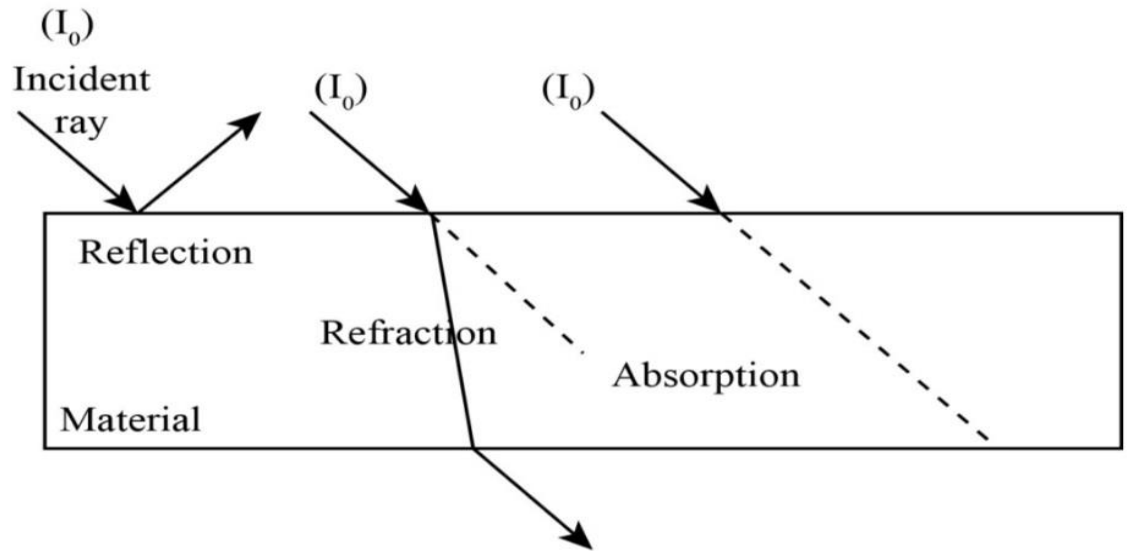
Interaction of light with a solid

The intensity of light wave represented by the given expression:

$$I_0 = I_t + I_a + I_\ell + I_r$$

where I_0 and I_t represent the intensity of incident and transmitted light respectively

I_a , I_ℓ and I_r represent the intensity of absorbed, reflected and refracted wave



Representation of reflection, refraction and absorption

Types of optical properties of materials

Different materials show different optical properties.

*Reflectivity

When a light wave is incident on the polished surface of the material, the light is reflected from the surface. Some materials show better reflectivity compared to other materials.

White surface shows better reflection of light

*Refractivity

When a light ray moves from one medium to another, the path of incident ray changes after the interaction with another medium. This property of material depends on the value of the refractive index of the material.

***Transparency**

An object can be seen clearly through some important transparent materials *like glass, air, water, etc.*

***Translucent**

An object can not be seen clearly through some materials *like sunglasses, vegetable oils, etc.*

***Opaque**

Some materials do not allow the passage of light waves. This means light waves can not pass through them. Examples: *wood, sand, metals*, etc. The application of this type of material prevents the transmission of light.

***Absorptivity**

When a light wave is incident on the polished surface of the material, some parts are absorbed by the material. ***Black surface shows better absorption of light.***

The absorption process depends on the frequency of the radiation and on the substance involved.

1- The Fundamental Absorption

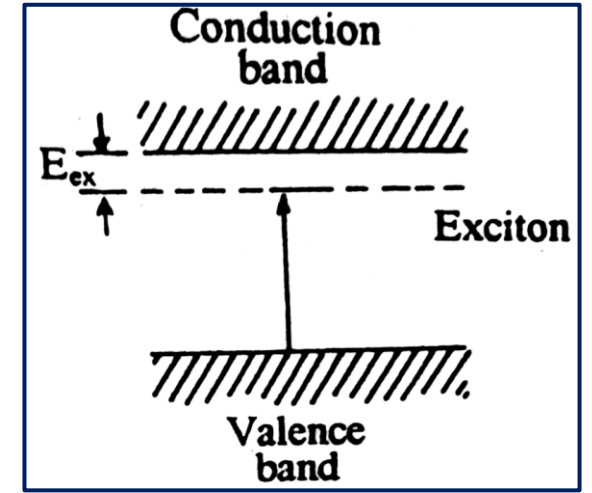
An electronic transition takes place between the occupied valence band and the unoccupied conduction band. An electron in the valence band absorbs a photon of energy hf and jumps into the conduction band,

$$hf \geq E_g \quad \text{or} \quad f \geq (E_g/h)$$

2- Exciton Absorption

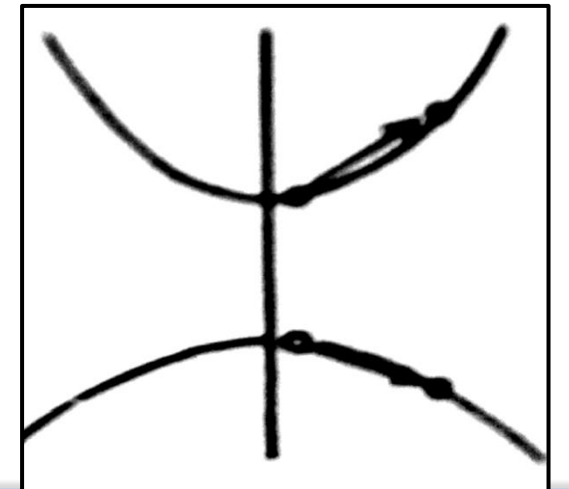
An electron might be excited such that it remains in the vicinity of its nucleus which in turn forms an electron-hole bound state. An electrostatic attraction exists between them, and the electron-hole pair is called an exciton.

The exciton binding energy is about $E_{\text{ex}} = 0.01$ eV. The exciton level are in the energy gap as in figure. An exciton introduces complications into the determination of E_g in semiconductors.



3-Free-carrier Absorption (Inter-band transition)

Free carriers (electrons and holes) absorb radiation and make transitions into another energy states in the same band. Free-carrier absorptions can take place even if $hf < E_g$. However, for $hf > E_g$, both fundamental absorption and free-carrier absorption can occur simultaneously.



***Luminescence**

It is the process in which a material absorbs energy and then immediately emits visible or near visible radiation. This phenomenon occurs due to excitation of electrons of material from valence band to the conduction band.

***Photoluminescence**

When the luminescence is produced by the bombardment of photon of an electromagnetic radiation lying in the range from **infrared to X-rays**.

***Fluorescence**

The luminescence in which the time interval between the acts of excitation and emission is the same.

***Phosphorescence**

The luminescence process in which luminescent emission continues for some time (perhaps hours) after the excitation caused. Example, when a crystalline ZnS illuminated by X-rays

***What is the importance of optical properties?**

It is very important in various industrial, scientific applications,.....

Photoconductivity

It is the increase in the electrical conductivity due to the production of electron –hole pairs by the incident light, photons. This phenomenon is used in light meters (in cameras) and solid-state infrared detectors.

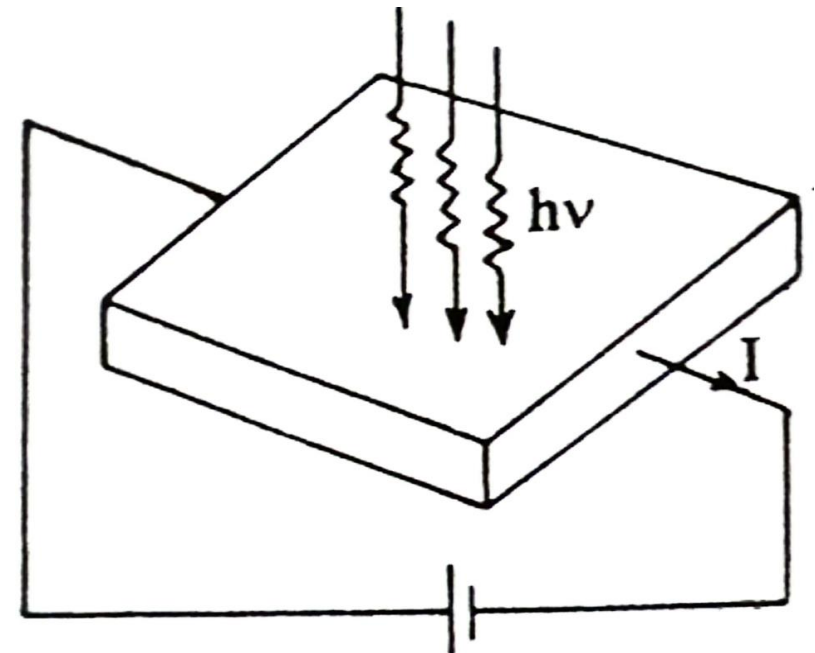
A simple experimental setup is shown in figure.

In dark, a semiconductor slab shows a negligible current flows, the dark conductivity is written as

$$\sigma_{\text{dark}} = ne\mu_e + pe\mu_h \approx 0$$

where **n** and **p** are free electrons and hole concentrations.

μ_e and **μ_h** are the mobilities of the electron and hole.



When the electromagnetic radiation of an appropriate frequency is allowed to fall on the semiconductor, the electrical conductivity increases. The change in the conductivity is given by:

$$\Delta\sigma = \sigma - \sigma_{\text{dark}} = \Delta n e (\mu_e + \mu_h)$$

$$\Delta n = f\tau$$

where τ is the carrier lifetime and f is the rate of their generation . (i.e. f : number of electron-hole pairs production per second by the absorbed photons).

In a practical device, the sensitivity is expressed as a gain, G .

G : It is the ratio of the carrier photocurrent, I , to the rate of generation of carriers. $G = I/ef$

Thus, Photoconductivity is large when τ is large. A pure material is generally found to have low τ (10^{-6} s) due to the greater probability of electron-hole recombination, and hence are relatively insensitive. However, a careful doping can suppress recombination and hence increase τ and sensitivity. For example, in CdS careful doping of iodine increases τ ($\approx 10^{-3}$ S).

Nd: YAG Laser (Doped insulator laser)

Lasing medium

What are the essential components of any laser system ?

The host medium is Yttrium Aluminium Garnet (YAG = $\text{Y}_3\text{Al}_5\text{O}_{12}$) with 1.5% trivalent neodymium ions (Nd^{3+}) as impurities. The (Nd^{3+}) ions occupy the lattice sites of yttrium ions as substitutional impurities and *provide the energy levels for both pumping and lasing transitions.*

Pumping source

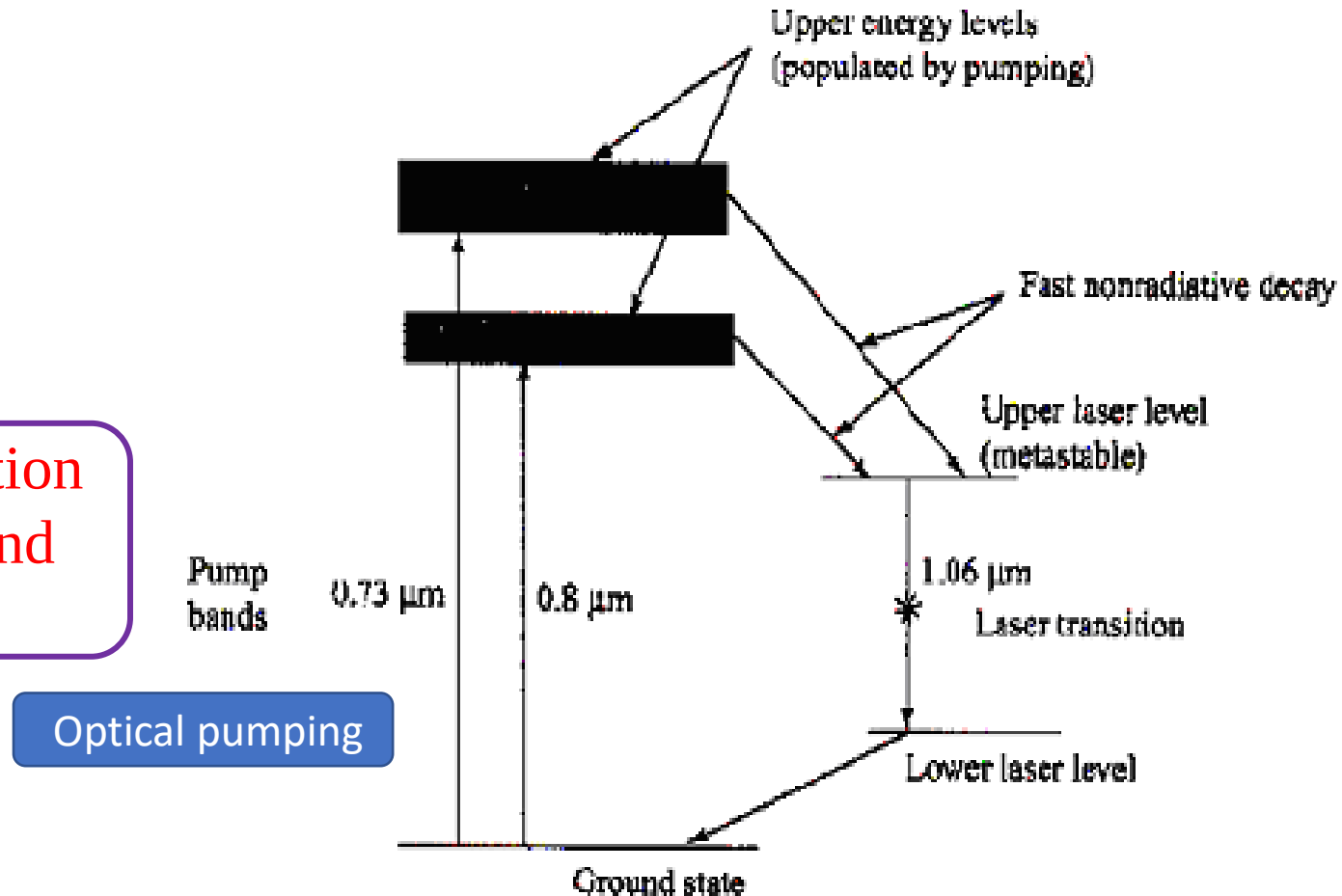
Flashlamp (*Laser diodes*) are now commonly used as the pump source owing *to their high efficiency* and good match to the absorption bands of Nd:YAG.

Optical resonator

The optical cavity is formed either by silvering the two ends of the rod or by external reflecting mirrors. One mirror is made hundred percent reflecting while the other mirror is left slightly transmitting to draw the output laser.

Simplified energy level diagram for the Nd-ion in YAG showing the principal laser transitions

It has two absorption bands ($0.73\ \mu\text{m}$ and $0.8\ \mu\text{m}$)



Nd:YAG laser applications

1- Scientific applications

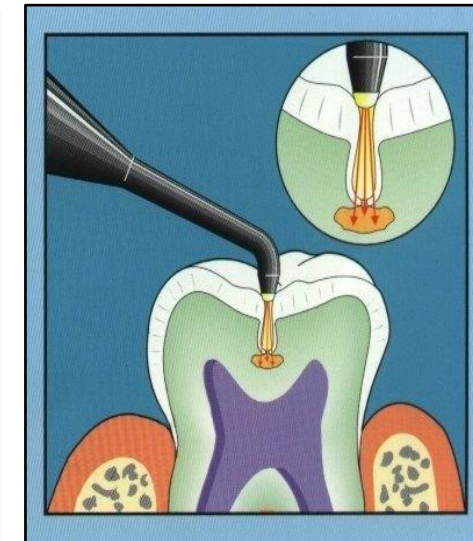
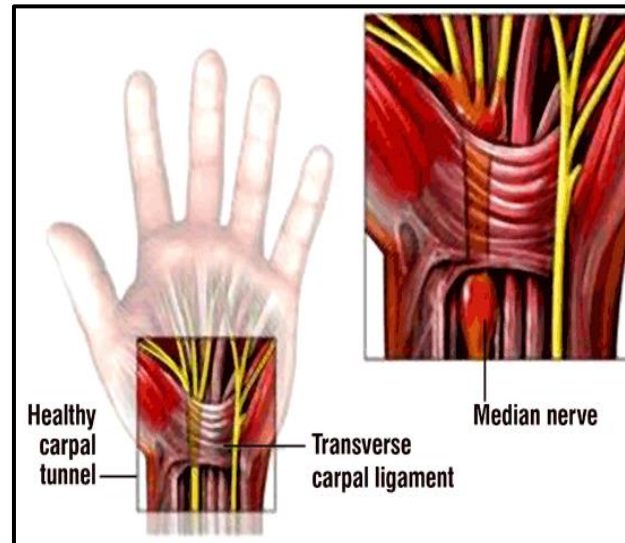
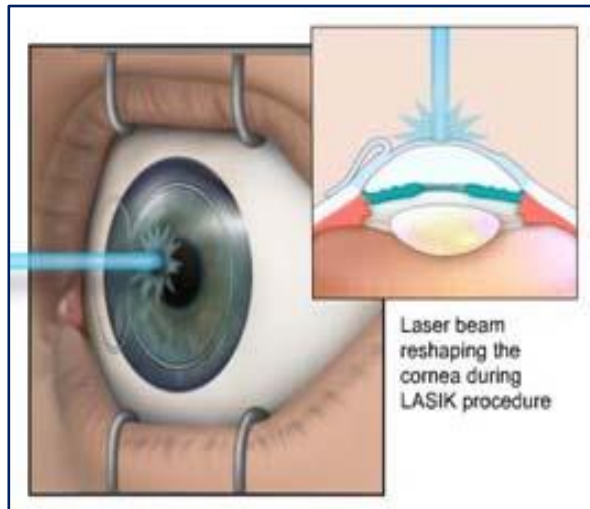
Research – Pumping other lasers to produce a green 532 nm beam

One of the most common high power lasers. Usually pulsed (down to fractions of a [nanosecond](#)) The Nd:YAG is very common as a pump laser (for e.g., dye lasers) and as a source of high-power pulses in the visible and ultraviolet regions.

2- Industrial use in materials processing such as welding, cutting, drilling.

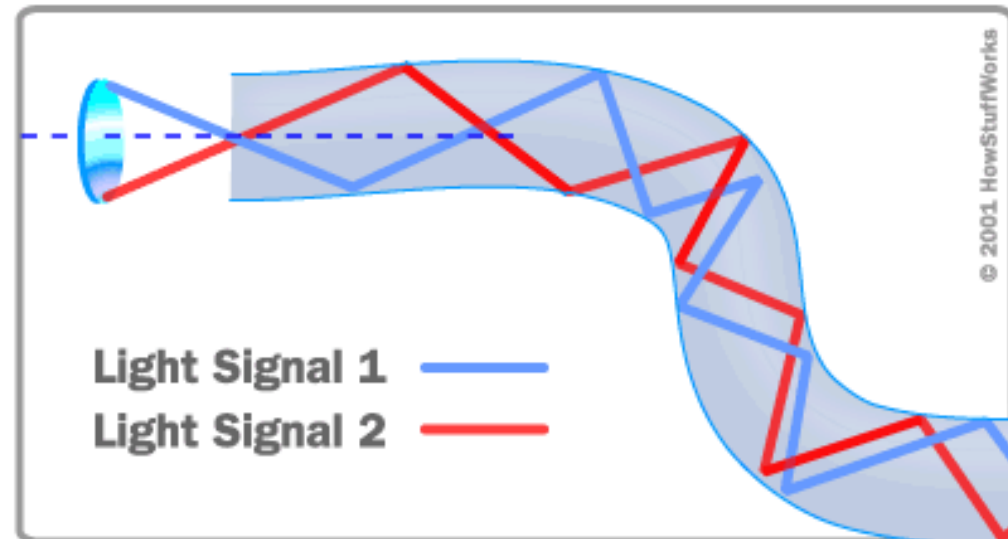
3- In Medicine: Surgery

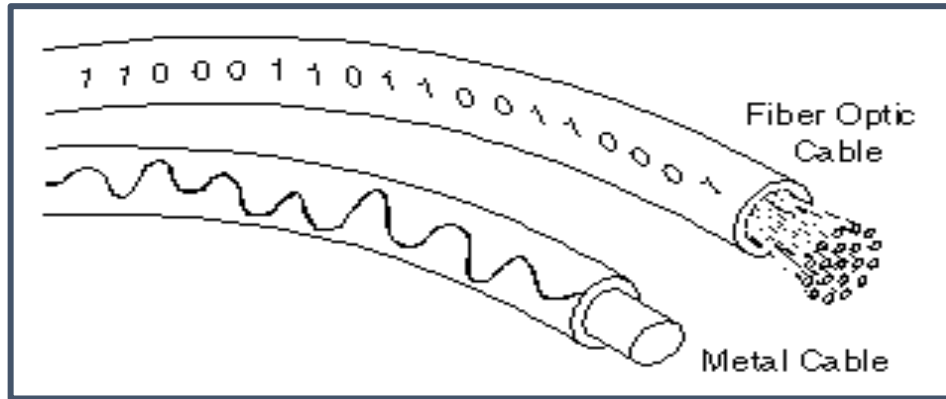
- a) Since $1.06\text{ }\mu\text{m}$ radiation passes through optical fibre without absorption, fibre optical endoscopes with YAG lasers are used to treat gastrointestinal bleeding.
- b) YAG beams penetrate the lens of the eye to perform intracocular procedures.



Communications

Lasers and fiber optics will soon replace most wires





A single strand of glass optical fiber can carry **more than half a million telephone conversations, or thousands of computer connections and TV channels.**

Thinner than a human hair, fiber optics promised unlimited expansion for the future.

Fiber optics has been the driving force in the enormous growth of the internet, the single most influential social and technological development of the 20th century... a development that was made possible by the semiconductor laser.